

Bora Girgin

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SUMMARY

Electrical and Electronics Engineering student focused on embedded systems, IoT and hardware-software integration. Experienced in designing and implementing real-world systems using ESP32, STM32 and Raspberry Pi, including RF communication (LoRa), PCB design and embedded Linux systems. Strong interest in building scalable, secure and production-ready embedded infrastructures.

EDUCATION

Sakarya University

BSc. in Electrical and Electronics Engineering

Sakarya, Türkiye

September 2022 – Present

Biga İCDAS Science High School

Biga, Canakkale

September 2018 – June 2022

PROJECTS

Raspberry Pi Monitoring & Control Dashboard

2025 – Present

Embedded Linux System

FastAPI, React, Raspberry Pi, Networking

Designed and developed a full-stack monitoring and control system for Raspberry Pi devices

Implemented secure remote access using Tailscale VPN with non-public network architecture

Designed a real-time telemetry system for CPU, memory, disk and temperature monitoring

Developed service management, terminal access and alert system for remote device control

LoRa Automatic Control System

November 2024 – Present

IoT Control System

LoRa, C++, ESP32, Raspberry Pi

Built an industrial automation system using long-range LoRa communication

Established reliable communication between ESP32 nodes and Raspberry Pi gateway

Achieved stable long-range communication (¿1km) in real-world testing

Developed as part of a TÜBİTAK-funded project in collaboration with Talu Textile

LoRa RF PCB Design

2024 - Present

Hardware Design Project

KiCad, RF Circuits, Embedded Systems

Designed a custom LoRa PCB using SX1276 transceiver

Optimized PCB layout for RF signal integrity and stable communication

Implemented ESD protection and power regulation circuits

Bluetooth Arduino Car Project

April 2020 – May 2020

Robotics Project

Bluetooth, IR, Ultrasonic, LIDAR

Developed a robotic vehicle with autonomous movement and remote control

Integrated IR, ultrasonic and LIDAR sensors for navigation and obstacle detection

Designed mechanical arm system using electric motors and FSR sensors

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Rust (basic), C#

Embedded & IoT: ESP32, STM32, Raspberry Pi, Arduino, LoRa, Embedded C/C++, Sensor Integration

Systems & Networking: Linux, Bash, TCP/IP, IoT Systems

Engineering Tools: MATLAB, Simulink, KiCad, Proteus, AutoCAD

Software & Tools: Git, SQL (MS SQL, SSMS)